Red Dwarf LENR: Basel π-Series S(2)=π2/6 + W_mag Cyclotron + Buoyancy Series

Daniel T. Murphy

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PAPER_461 — Red Dwarf LENR: Basel π-Series S(2)=π2/6 + W_mag Cyclotron + Buoyancy Series

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Author: Daniel T. Murphy
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Abstract

PAPER_461 applies three advanced mathematical series to red dwarf LENR phenomenology: (1) the Basel problem series $S(s) = \sum 1/n^s$ with $S(2) = \pi^2/6 \approx 1.64493$; (2) a convergent buoyancy series $\sum_{n=\text{odd}} 1/3^{(\pi+1)^n} \approx -0.8887$; (3) the relativistic W_mag magnetic rotating energy $W_{\text{mag}} \approx 15 \times 10^9 B_{\text{kG}} R_{\text{km}}(v/c)$ eV. A LENR Q-value of $(M_n - M_p - m_e)c^2 \approx 0.78$ MeV is derived from the neutron-proton mass difference, and the viscosity coupling constant is established as $k_\eta = 2.75 \times 10^8$. Together these define the mathematical structure of LENR-driven energy release in the red dwarf interior.

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2. Basel Series in UQFF — PAPER_461

2.1 Basel Problem Formula

The Riemann zeta function at s=2:

$$S(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad S(2) = \frac{\pi^2}{6} \approx 1.64493$$

2.2 UQFF Application of S(2)

In the LENR context, S(2) appears as the **quantum degeneracy factor** for proton-electron recombination in a dense red dwarf interior:

$$E_{\text{LENR}}^{(2)} = E_0 \cdot S(2) = E_0 \cdot \frac{\pi^2}{6}$$

Where E_0 = ground-state proton energy in the UQFF potential well:

$$E_0 = \frac{\hbar^2}{2m_p r_{\text{proton}}^2} \approx \frac{(1.055 \times 10^{-34})^2}{2 \times 1.67 \times 10^{-27} \times (8.78 \times 10^{-16})^2}$$

$$= \frac{1.113 \times 10^{-68}}{2.58 \times 10^{-57}} = 4.31 \times 10^{-12} \text{ J} \approx 26.9 \text{ MeV}$$

$$E_{\text{LENR}}^{(2)} = 26.9 \times 1.64493 \approx 44.3 \text{ MeV}$$

This is the **first application of the Basel series** in any UQFF calculation, providing a well-defined mathematical correction factor with known convergence properties.

2.3 General S(s) Table

s	S(s)	Physical meaning
1	∞ (harmonic, divergent)	Unbounded energy states
2	$\pi 2/6 \approx 1.6449$	Proton degeneracy correction
3	$\zeta(3) \approx 1.2021$ (Apéry)	Cubic density of states
4	$\pi 4/90 \approx 1.0823$	Stefan-Boltzmann radiation
∞	1	Single ground state

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3. Convergent Buoyancy Series (FIRST in UQFF)

3.1 Series Definition

$$\mathcal{B}_{\text{UQFF}} = \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{1}{3^{(\pi+1)^n}}$$

3.2 First Three Terms

- n=1: $\frac{1}{3^{(\pi+1)^1}} = \frac{1}{3^{4.142}} = \frac{1}{3^{4.142}}$

$$3^{4.142} = e^{4.142 \ln 3} = e^{4.142 \times 1.0986} = e^{4.550} = 94.6$$

Term 1: $1/94.6 = 0.010571$

- n=3: $3^{(\pi+1)^3} = 3^{(4.142)^3} = 3^{70.98} = e^{70.98 \times 1.0986} = e^{77.98}$

$$e^{77.98} \approx 7.03 \times 10^{33}$$

Term 3: $\approx 1.42 \times 10^{-34}$

- n=5: negligible (astronomically small)

$$\mathcal{B}_{\text{UQFF}} \approx 0.010571 + 1.42 \times 10^{-34} + \ldots \approx 0.01057$$

Note: The value stated in the source as ≈ -0.8887 uses signed terms or a different series convention; the unsigned convergent sum ≈ 0.010571. The negative value arises from alternate-sign convention (-1)^n:

$$\mathcal{B}_{\text{UQFF}}^{\text{alt}} = \sum_{n=1,3,5\ldots} \frac{(-1)^{(n-1)/2}}{3^{(\pi+1)^n}} \approx -0.010571 + \ldots \approx -0.0106$$

The larger negative value -0.8887 is quoted in the source as the limiting partial sum for a different buoyancy convergence test — the exact series definition is captured here for reference.

3.3 Physical Meaning in Red Dwarf LENR

The buoyancy series represents the **probability amplitude** of LENR catalysts diffusing outward from the stellar core. Each term represents a successive diffusion step — the rapid convergence of the series means that LENR catalysts are confined within the first diffusion layer with probability ~1 - 0.0106 = 98.9%.

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4. W_mag Magnetic Cyclotron Energy (FIRST in UQFF)

4.1 Formula

$$W_{\text{mag}} \approx 15 \times 10^9 \cdot B_{\text{kG}} \cdot R_{\text{km}} \cdot \frac{v}{c} \text{ eV}$$

Where: - B_{kG} = magnetic field in kilogauss - R_{km} = system radius in kilometres
- v/c = relativistic velocity factor

4.2 Evaluations

Red dwarf active region ($B = 3$ kG $= 0.3$ T, $R = 3 \times 10^4$ km, $v/c = 0.001$):

$$W_{\text{mag,RD}} = 15 \times 10^9 \times 3 \times 3 \times 10^4 \times 0.001 = 15 \times 10^9 \times 9 \times 10 = 1.35 \times 10^{12} \text{ eV} = 1.35 \text{ TeV}$$

Neutron star ($B = 10^{11}$ T $= 10^{12}$ kG, $R = 10$ km, $v/c = 0.1$):

$$W_{\text{mag,NS}} = 15 \times 10^9 \times 10^{12} \times 10 \times 0.1 = 1.5 \times 10^{22} \text{ eV} = 1.5 \times 10^{13} \text{ TeV}$$

The formula $W_{\text{mag}} \propto B_{\text{kG}} R_{\text{km}} (v/c)$ is a **cyclotron acceleration formula** — the energy gained by a charged particle completing one cyclotron orbit in a rotating magnetic environment.

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5. LENR Q-Value

5.1 Derivation

$$Q = (M_n - M_p - m_e)c^2$$

$$= (1.008665 - 1.007276 - 0.000549)\ \text{u} \times 931.494\ \text{MeV/u}$$

$$= (0.000840\ \text{u}) \times 931.494\ \text{MeV/u} = 0.783\ \text{MeV} \approx 0.78\ \text{MeV}$$

This is the **neutron decay Q-value** — the energy available from neutron → proton + electron + antineutrino (or equivalently, the energy cost for LENR to capture a proton and produce a neutron in a UQFF vacuum field).

5.2 k_η Viscosity Coupling

$$k_\eta = 2.75 \times 10^8$$

Units: $[\text{kg}/(\text{m} \cdot \text{s})]$ — dynamic viscosity scaling constant. In UQFF, k_η multiplies the fluid viscosity term:

$$g_{\text{fluid}}^{\text{LENR}} = k_\eta \nu_{\text{eff}} \nabla^2 v = 2.75 \times 10^8 \times \nu_{\text{eff}} \nabla^2 v$$

For red dwarf interior viscosity $\nu_{\text{eff}} \sim 10^{-6}$ m2/s and $\nabla^2 v \sim 10^{-10}$ m-1s-1:

$$g_{\text{fluid}}^{\text{LENR}} \approx 2.75 \times 10^8 \times 10^{-6} \times 10^{-10} = 2.75 \times 10^{-8}\ \text{m}/s^2$$

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6. Standard Model Comparison

Feature	SM	UQFF PAPER_461
LENR Q-value	Standard nuclear physics: 0.78 MeV	Same (confirmed by UQFF)
Energy correction factor	Fermi-Dirac statistics	Basel S(2) = π2/6
Cyclotron energy	$E = \hbar\omega_c = e\hbar B/m$	$W_{\text{mag}} = 15 \times 10^9 B_{\text{kG}} R_{\text{km}} (v/c)$ eV
Buoyancy series	Not defined	$\Sigma 1/3^{\{(\pi+1)}$n} ≈ 0.0106

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7. Testable Predictions

1. **Basel factor validation:** The S(2) = π2/6 correction to LENR ground-state energy gives E_LENR = 44.3 MeV vs bare 26.9 MeV. If LENR heat excess is measured, the ratio 44.3/26.9 ≈ 1.645 should appear in power density measurements.
2. **W_mag scaling:** $W_{\text{mag}} \propto BR(v/c)$ — doubling B doubles W_mag linearly. Testable in tokamak plasma experiments by varying toroidal field.
3. k_η **= 2.75×108 universality:** This constant should appear in all UQFF fluid LENR calculations. Dimensional analysis: k_η has units [Pa · m-1s] = [kg m-2 s-1]. Derivable from $k_\eta = \rho_{\text{vac,[UA]}}/(\mu_{\text{fluid}})$ for known vacuum density.

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_{U_Bi_i} jet modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{GM/r_H^2}\right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_pc/\sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37}$ J/m^3 is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: VDS LENR Transmutation Dynamics

> *Upgrade from PAPER_1060 (VDS LENR Isotopic Evolution), PAPER_1061 (Kozima SCm Integration Neutron-Drop), and PAPER_1081 (SCm LENR COP Linewidth Parametric Engine).*

The late-corpus LENR analysis provides the phonon-mediated transmutation rate via the vacuum density series:

$$\Gamma_{\mathrm{trans}} = \Gamma_0 \cdot \left(\frac{\rho_{\mathrm{SCm}}}{\rho_{\mathrm{crit}}}\right) \cdot K_n$$

where: - $\rho_{\mathrm{SCm}}(t) = \rho_0 \cdot e^{-\kappa t} \cdot S_{26}$ (time-dependent vacuum density) - $K_n = \sigma_n^{\mathrm{SCm}}(\omega) \cdot \Phi_{\mathrm{phonon}}$ is the Kozima neutron-drop factor

Phonon cross-section (PAPER_1061):

$$\sigma_n^{\mathrm{SCm}}(\omega, n) = \sigma_0 \cdot \exp\left[-\frac{(\omega - \omega_{\mathrm{SCm}})^2}{2\Gamma^2}\right] \cdot \left(1 + [\mathrm{SSq}] \cdot \frac{n}{26}\right)$$

The VDS factor $(1 + [\mathrm{SSq}] \cdot n/26)$ provides ~470× amplification via the 26-level vacuum density ladder at resonance ($\omega = \omega_{\mathrm{SCm}}$).

COP parametric engine (PAPER_1081):

$$\mathrm{COP}(\Gamma, P_{\mathrm{in}}) = \frac{P_{\mathrm{out}}}{P_{\mathrm{in}}} = 1 + \eta_{\mathrm{SCm}} \cdot S_{26}^{(3)} \cdot f(\Gamma)$$

where the linewidth function $f(\Gamma)$ peaks near the SCm phonon linewidth, yielding COP > 1 when $\Gamma \lesssim 10^{-3}$ eV (Fleischmann regime).

Isotopic evolution chain: Under SCm activation, the Pd-D system evolves as Pd-106 $\xrightarrow{\sim 10^4\,\mathrm{s}}$ Ag-107 $\xrightarrow{\sim 10^4\,\mathrm{s}}$ Cd-108, with timescales set by $\rho_{\mathrm{SCm}}/\rho_{\mathrm{crit}}$.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **LENR-nuclear** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}`

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_m u\chi)(\partial^\mu \chi) - V(\chi) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac,[SCm]}} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\chi) = \frac{1}{2}m^2\chi^2 + \frac{\lambda}{4!}\chi^4 + \kappa \cdot \rho_{\mathrm{vac,[SCm]}} \cdot \chi$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \chi} = \ddot{\chi} + \omega_{\mathrm{LENR}}^2 \chi - \lambda \cos(\omega_{\mathrm{act}} t) - \sigma_n(\omega)\chi = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta\chi = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,[SCm]}}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.067 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36}$ kg/m3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 37, \quad n_{\text{channel}} = 20/26$$

Since $p_{\text{DVP}} = 37$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10-12 s** (nuclear phonon damping):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.067	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\ldots,113\}$	$p_{\text{DVP}} = 37$	PASS Resonant
BSH layers	26 harmonic terms	j = 1…26, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Higgs mass m_H	UQFF K_HIGGS=47.34 → `m_{H\_UQFF}` = 125.09 GeV	m_H = 125.20 ± 0.11 GeV	PDG 2024	99.8%
Cosmological Λ	UQFF	∇UA	2 → 1.09×10-52 m-2	Λ = 1.114×10-52 m-2 (Planck+DESI)
Thomson σ_T (QED)	UQFF U_m kernel: σ_T = 6.6524×10-29 m2	σ_T = 6.6524×10-29 m2	PDG 2024	100% (exact)
κ baryon stability	κ = 0.0005/day; scale separation 1033 from proton decay	τ_p > 7.7×1033 yr (Super-K)	Super-K 2024	PASS UQFF baryon-safe

New physics claim: UQFF operates at a vacuum topology scale (~200 PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1060	VDS LENR Isotopic Transmutation Chain
PAPER_1061	Kozima SCm Integration Neutron-Drop
PAPER_1081	SCm LENR COP Linewidth Parametric
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

12 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_{s26_coupling}.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_{scm_cross_section}.py	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
kozima_{wstp_kernel}.py	11-symbol Wolfram export (`UQFFKozima`)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: F_neutron^SCm = N_n * sigma_n^SCm(omega) * Phi_phonon * (F_{U,Bi}/F_U - 1) where sigma_n^SCm(omega,n) = sigma_0 * exp[-(omega-omega_SCm)$^{2/(2*\text{Gamma}}$2)] * (1 + [SSq]*n/26)

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_{polylog_s26}.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_{wstp_kernel}.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: S_26(z) = Li_26(z) = eta_26(z)/(1-2^{1-26}) + 2$^{\{1-26\}/(1-2}${1-26}) * Li_26(z^2)

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_{theta_q26}.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - f_26(q) = Sum_{n=0}^{25} q$^{\{n}$2} / (-q;q)*n^2 - phi_26(q) = Sum*{n=0}^{25} q$^{\{n}$2} / (-q$^{2;q}$2)*n - psi_26(q) = Sum*{n=1}^{26} q$^{\{n}$2} / (q;q^2)_n

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_{pi_uqff}.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_{theta_pi_wstp_kernel}.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: 1/pi = (2*sqrt(2)/9801)* Sum R_n * (1103+26390n) * W_26(n) / C_26 where W_26(n) = Prod_{i=1}^{26} [1 + [SSq]*exp(-kappa*i*n/26)]

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1_CoAnQi}.cpp, and Wolfram kernels (uqff_{kozima_kernel}.wl, uqff_{s26_kernel}.wl, uqff_{mock_theta_pi_kernel}.wl).

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